

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110013**Date: 16-01-2019****Subject Name: ENGINEERING GRAPHICS & DESIGN****Time: 10:30 am to 01:00 pm****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		Marks
Q.1	(a) For 100 cm of a line compare size of drawing length on basis of full scale, reducing scale & enlarged scale.	03
	(b) Prepare isometric scale to measure 40mm and 74mm.	04
	(c) Point P of a straight line PQ is 25mm above H.P. and point Q is 65 mm in-front of V.P. The line makes an angle of 30° with H.P. and its plan is at 45° to the XY line. Draw the projections of the line if the plan length is 70mm. Also find the true length of the line and the angle made by the line with V.P.	07
Q.2	(a) List two applications of an ellipse, parabola and Hyperbola.	03
	(b) For the cuboid 50 x 30 x 20, draw all six orthographic views using 3 rd angle projection method.	04
	(c) A point P moves towards another point O, 90 mm from it, and reaches it during 1.5 revolutions around it in clockwise direction. Its movement towards O is uniform with its movement around it. Draw the curve traced out by the point P and name it.	07
	OR	
	(c) A fixed point is 54 mm away from a fixed straight line. Draw the locus of a point P moving in such a way that the ratio of its distance from the fixed straight line is 5:4. Name the curve.	07
Q.3	(a) Draw projection of following points (i) Point R is 10 mm behind V.P. & 20 mm above H.P. (ii) Point S is in H.P. & 22 mm in front of V.P. (iii) Point T is 15 mm in front of V.P & 25 mm below H.P.	03
	(b) A line AB, 75mm long, is parallel to VP and inclined to the HP, by an angle 45° . Point A is 30mm below HP and 20mm in front of VP. Point B is in the first quadrant. Draw the projections of the straight line AB.	04
	(c) A line PQ, 100 mm long, is inclined at 30° to the HP and 45° to the VP. Its mid-point M is in the VP and 20mm above the HP. Draw its projections, when its end P is in the first quadrant and Q is in the third quadrant.	07
	OR	
Q.3	(a) Draw Projections of the following lines. (i) Line MN 50mm is in 1 st quadrant and parallels both H.P. & V.P. (ii) Line PQ 35mm is in 3 rd quadrant and remains perpendicular to V.P. and parallel to H.P.	03
	(b) A square plate PQRS, edge 25mm, is in space with one of its corners in VP. Surface of the plate makes 50° with VP and it is perpendicular to HP. Draw its projections.	04
	(c) The distance between the end projectors of a straight line PQ is 60mm. The line makes 30° and 45° angles with HP and VP, respectively. The	07

end P is 30mm below HP and 50mm in front of the VP. Draw its projections when end Q is in third quadrant. Find TL of the line.

- Q.4** (a) Define apparent shape and true shape with diagram. **03**
 (b) Draw the development of pentagonal prism of side 30mm and height 60mm, when one of the edges of the base is perpendicular to VP. **04**
 (c) An isosceles triangular plate ABC has its base 45mm and altitude 60mm. **07**
 It is so placed that the front view is seen as an equilateral triangle of 45mm side and (i) base is inclined at 45° to HP (ii) side is inclined at 45° to HP. Draw its plan when its corner A is on HP.

OR

- Q.4** (a) Define right solid, oblique solid and regular solid. **03**
 (b) A cone, diameter of base 55mm and height 60mm, is resting on HP on one of its generators with axis parallel to VP. Draw the projections of cone. **04**
 (c) A semicircular plate of 80mm diameter has its straight edge in the VP and inclined at 45° to the HP. The surface of the plate makes an angle of 30° with the VP. Draw its projections. **07**
- Q.5** (a) Why chamfer is done on work piece. Write the steps to create chamfer in AUTOCAD. **03**
 (b) List and explain different methods to draw circle in AUTOCAD. **04**
 (c) Draw the (i) Front view (ii) Right hand side view and (iii) Top view of Fig. 02 in first angle projection method. Consider length as 50mm in direction of X. **07**

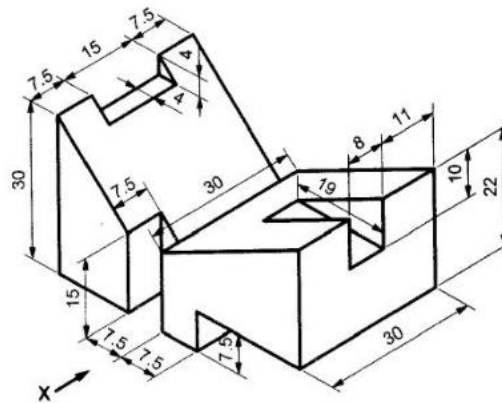


Fig.02

OR

- Q.5** (a) Write difference between line, polyline and its uses in AUTOCAD. **03**
 (b) List and explain different methods to draw rectangle in AUTOCAD. **04**
 (c) Draw isometric view of the Fig.03 given below. **07**

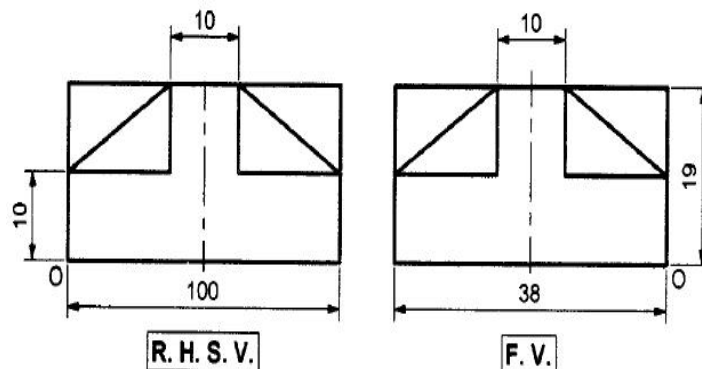


Fig.03

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I &II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110013****Date: 18/06/2019****Subject Name: Engineering Graphics & Design****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1** (a) Draw a diagonal scale of R.F, 1:5 showing decimeters, centimeters and millimeters and long enough to measure upto 8 decimeters. Show a distance of 5.35 dm. **03**
- (b) In the mechanism shown in Fig.1, the connecting rod is constrained to pass through the trunnion at D. Trace the locus of the end C and a point P on BC for one complete revolution of the crank. In Fig. 1 consider AB as 30 mm. **04**
- (c) A line AB is 100 mm long. It is inclined at 40° to the HP and 30° to the VP. The end A is 10 mm above HP and 25 mm in front of VP. Assuming the end B in the first quadrant, draw the projections of the line AB. **07**
- Q.2** (a) Explain the following autocad commands : hatch, circle and array. **03**
- (b) Explain the following autocad commands : mirror, trim, extend and fillet. **04**
- (c) A triangle ABC has sides AB = 75 mm, BC = 60 mm and CA = 75 mm. Draw a parabola passing through points A, B and C when side BC is horizontal. **07**
- OR**
- (c) Draw an epicycloid with rolling circle diameter 50 mm and directing circle diameter 150 mm. Draw tangent and normal at a point on the curve 110 mm from the centre of the directing circle. **07**
- Q.3** (a) Define the following curves : involute, ellipse and cycloid. **03**
- (b) The distance between the end projectors of a straight line AB is 60 mm. Point A is 5 mm above HP and 30 mm in front of VP. Point B is 40 mm above HP and 50 mm behind VP. Draw the projections and find the inclination of straight line AB with HP and VP and the true length of the line. **04**
- (c) An isosceles triangular plate of 50 mm base and 75 mm altitude, appears as an equilateral triangle of 50 mm in top view. Draw the projections of a plate if its 50 mm long edge is on the HP and inclined at 45° to the VP. What is the inclination of the plate with the HP ? **07**
- OR**
- Q.3** (a) Draw the projections of the following points on the same x-y line : **03**
- (i) Point A is 20 mm above the HP and 20 mm behind the VP.
- (ii) Point B is 40 mm above HP and 10 in front of VP.
- (iii) Point C is 25 mm below HP and 40 mm behind VP.
- (b) Two lemons on a tree, planted near the compound wall of a bungalow are 1.0 m and 1.25 m above the ground and 0.5 m and 0.75 m from a 15 cm thick compound wall but on opposite sides of it. The distance between lemons measured along the ground and parallel to the wall is 1.0 m. Determine the real distance between the centres of two lemons. **04**

- (c) ABCDE is a regular pentagonal plate of 40 mm sides, has its corner A on the HP. The plate is inclined to the HP such that the plan length of the edges AB and AE is each 35 mm. The side CD is parallel to both the reference planes. Draw the projections of the plate and find its inclination with the HP. **07**
- Q.4** (a) Draw the projections of a cone, base 60 mm diameter and axis 90 mm long when it is resting on the HP on a point on its base circle with the axis making an angle of 30° with the HP and parallel to the VP. **03**
- (b) Fig.2 shows the front view of a cut hexagonal pyramid. Draw the development of the lateral surface of the remaining portion of the pyramid. **04**
- (c) A cylinder is resting on HP on its base. It is cut by AVP perpendicular to HP and inclined to VP by 45° and cutting it remaining 12 mm away from the axis. Draw the projections with section and draw also the true shape of the section. Take diameter of cylinder 55 mm and height 60 mm. **07**
- OR**
- Q.4** (a) Differentiate between **03**
- (i) Prism and pyramid
- (ii) Square pyramid and tetrahedron
- (iii) Cube and square prism
- (b) Draw the plan and elevation of a cone resting on HP on its base. Show on them the shortest path followed by a fly moving around the cone and returning to the same starting point. Fly starts from a point on the periphery of base. Take base diameter of cone 80 mm and height of axis 90 mm. **04**
- (c) A cone of 70 mm diameter of the base circle and 60 mm length of axis is resting on its base on the HP. It is cut by an AIP so that true shape of the section is an isosceles triangle with the vertex angle of 50° . Set the required cutting plane and find its inclination with the HP. Draw sectional top view, front view and project the true shape of the section. **07**
- Q.5** (a) Differentiate between aligned system and unidirectional system of dimensioning. **03**
- (b) Using third angle projection method draw right hand side view for the object shown in Fig. 3. **04**
- (c) Using third angle projection method draw front view and top view for the object shown in Fig. 3. **07**
- OR**
- Q.5** (a) A circular plate, 50 mm diameter is resting on HP on one of the points of its periphery with surface of the plate perpendicular to VP and inclined to HP by 30° . Draw two projections of the circular plate. **03**
- (b) Draw isometric view from the orthographic projection shown in Fig. 4. **11**

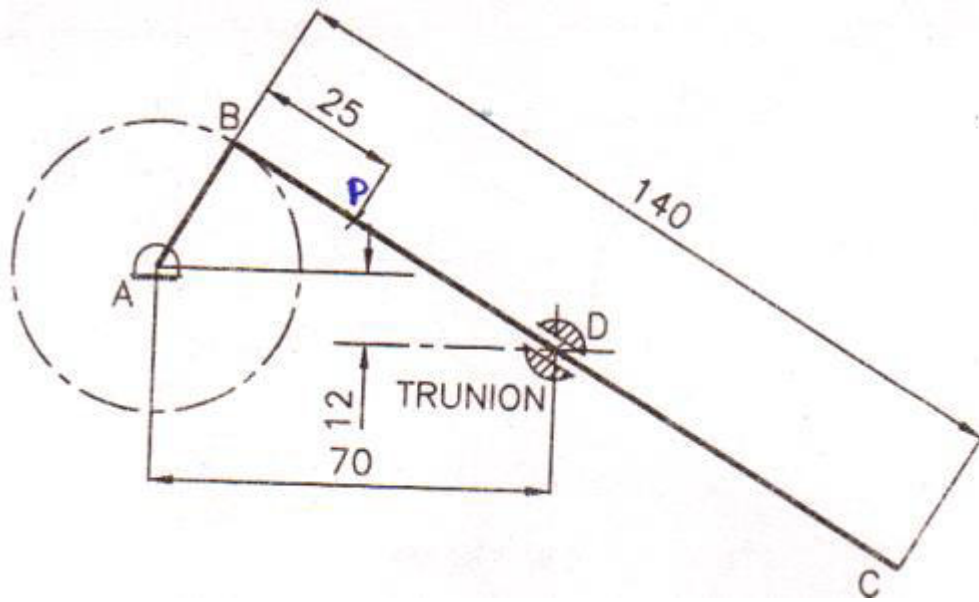


Fig. 1

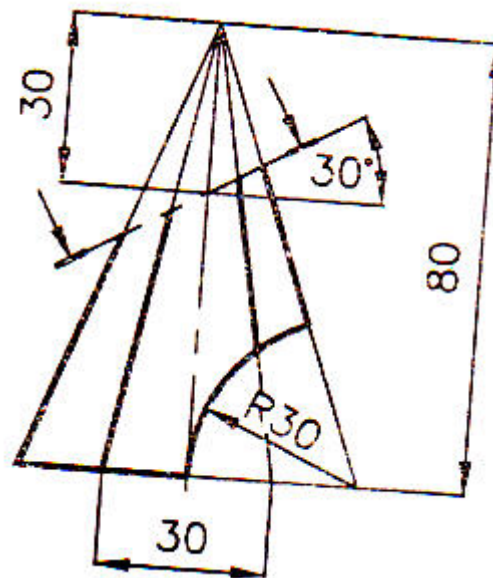


Fig.2

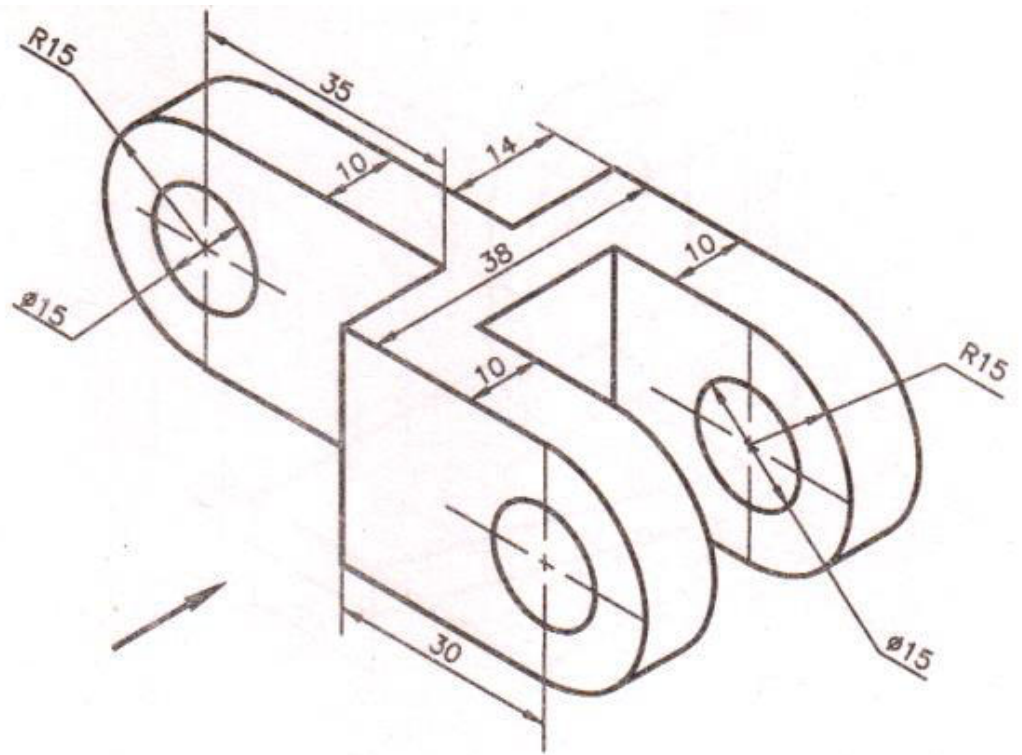


Fig.3

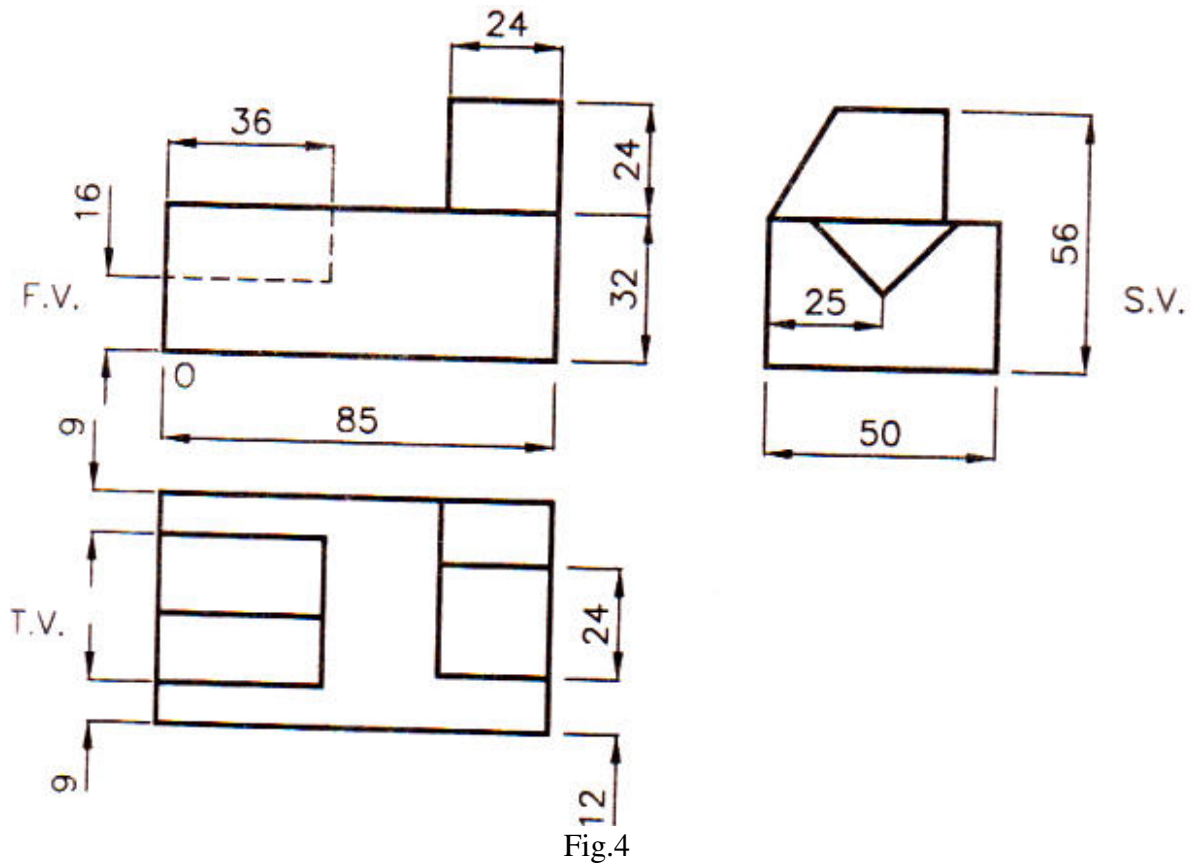


Fig.4

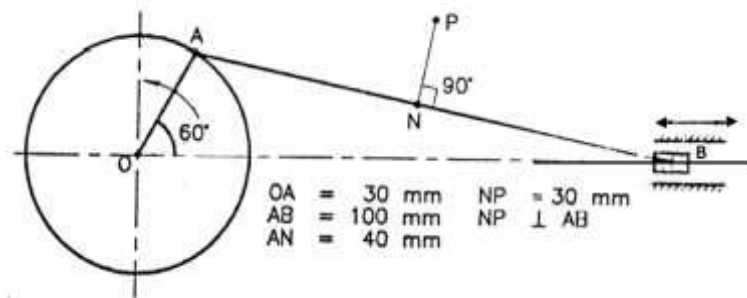
GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110013****Date: 09/01/2020****Subject Name: Engineering Graphics & Design****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Marks

- Q.1 (a)** Construct a scale of 1cm = 1 m to read meters and decimeters and long enough to measure up to 14 meters. Show on this scale a distance equal to 12.4 meters. **03**

- (b)** OAB is a slider crank mechanism. Slider B is sliding on a straight path passing through O as shown in Figure 1 given below. Crank OA is 30 mm and rotates in anticlockwise direction and length of connecting rod AB is 100 mm. A rod NP of 30 mm length is attached to AB such that AN = 40 mm and NP is perpendicular to AB as shown. Draw locus of point P for one complete revolution of the crank. **04**

**Figure 1.**

- (c)** Draw a parabola by tangent method, given its base as 90 mm and the height of axis is 75 mm. **07**
- Q.2 (a)** A rectangle ABCD 60 mm x 40 mm, is parallel to HP with one of its sides inclined at 30° to VP and the end of the side near to VP is 15 mm in front of the VP and 30 mm above HP. Draw its projections. **03**
- (b)** Point A is 20 mm above HP and 30 mm in front of VP and point B is in the HP and 40 mm behind the VP. The distance between in their projectors is 50 mm. Draw the projections of the points. Also draw straight lines joining their top and front views. **04**
- (c)** The end P of a line PQ 120 mm long is 30 mm above HP and 60 mm behind VP. The line is incline at angle of 30° with the reference plans of the projection. The point Q is below HP and behind VP. Draw the projections of line PQ and locate the point Q. **07**

OR

- (c)** Plan and elevation of a line AB, 80 mm long, measure 60 mm and 72 mm respectively. End A of the line is in HP and end B is in VP. Draw its projections, assuming the line to lie in first quadrant. **07**

- Q.3** (a) What is the difference between a First Angle and Third Angle projection? **03**
 (b) A right regular pentagonal pyramid, edges of base 25 mm and height 50 mm, has its base parallel to VP with one of its base edges in HP. Draw its projections. (FV and TV) **04**
 (c) A regular hexagon lamina ABCDEF has a side AB in VP and its side DE 45 mm in front of the VP and inclined to HP at 30° . Draw its projections. Consider size of lamina as 50mm. **07**

OR

- Q.3** (a) Draw the conventional symbols for First Angle and Third Angle projection. **03**
 (b) A right regular pentagon prism, edge of base 30 mm and height 75 mm resting on its base on HP, is cut by a section plane inclined to HP at 45° and meeting the axis at a distance of 18 mm from its top end. Develop the outside surface of the cut prism. **04**
 (c) Draw the projections of rhombus diagonals 96 mm and 48 mm long. The smaller diagonal is parallel to both the reference planes, while the other diagonal is inclined at 30° to the HP and has one of its end points in the HP. Keep the center of rhombus 56 mm in front of the VP. **07**

- Q.4** (a) Write difference between line, polyline and its uses in AUTOCAD. **03**
 (b) List and explain different methods to draw arc in AUTOCAD. **04**
 (c) One of the rectangular faces of the vertical square prism, 40 mm side and axis 65 mm long, make an angle of 30° to VP. It is cut by an AIP passing through the center of the axis in making an angle of 40° to HP. Draw front view, sectional top view and true shape of the section. **07**

OR

- Q.4** (a) List the six Essential Commands of Modify Panel in AutoCAD. **03**
 (b) List and explain different methods to draw Polygon in AUTOCAD. **04**
 (c) A square pyramid, side of base 40mm and axis 60 mm long, has its base in HP and all edges of the base are equally inclined to VP. It is cut by a section plane perpendicular to the VP and inclined 45° to the HP such that it bisect the axis. Draw its sectional top view and sectional left side view. **07**

- Q.5** (a) Draw the isometric scale of 80 mm long line and show 66 mm isometric length on it. **03**
 (b) Draw the isometric view of the cone of 48 mm base diameter and 56 mm axis height. **04**
 (c) Draw and dimension completely, the following views according to first angle projection method and to full size scale of a given figure 2. **07**
- Front view looking in the direction of arrow F,
 - Side view looking direction of arrow S,
 - Top view looking the direction of arrow P.

- (c) Figure 4 shows three orthographic views according to first angle projection method of an object. Draw its isometric projection. **07**

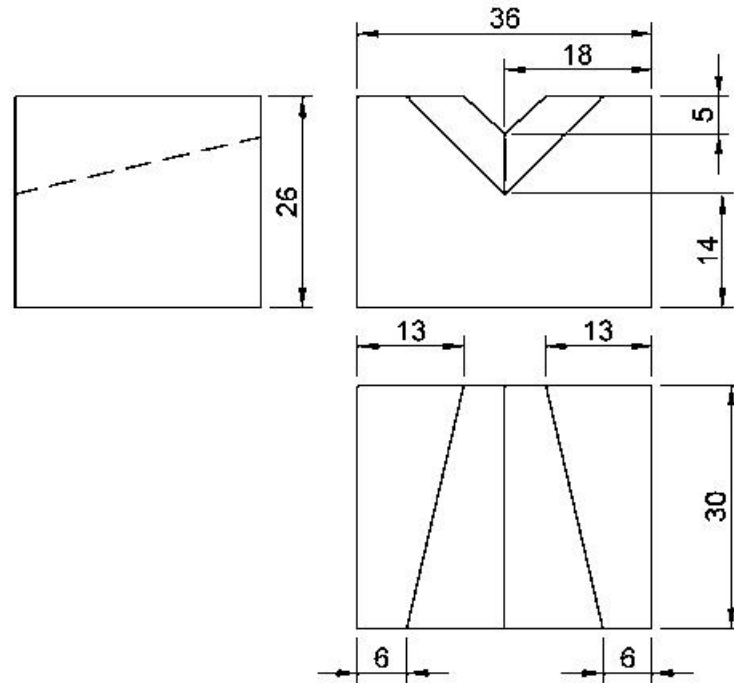


Figure 4.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE – SEMESTER 1&2 EXAMINATION – SUMMER 2020****Subject Code: 3110013****Date: 05/11/2020****Subject Name: Engineering Graphics and Design****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1** (a) Define Representative Fraction (RF). Find RF if 5 cm in the drawing is equal to 10 km. **03**
- (b) Give the definitions of the following curves: (i) Ellipse (ii) Involute **04**
- (c) Draw epicycloid of a 40 mm diameter circle, which rolls outside on another circle of 150 mm diameter for one revolution clockwise. Draw a tangent and normal to it at a point 95 mm from the center of the directing circle. **07**

- Q.2** (a) Draw the projections of the following points on the same X-Y line. **03**
- (a) Point 'A' is 20 mm below H.P and 20 mm in front of V.P.
- (b) Point 'B' is 30 mm above H.P and 40 mm in front of V.P.
- (c) Point 'C' is on V.P and 30 mm above H.P.
- (b) The top view of a 75 mm long line AB measures 65 mm, while its front view measures 50 mm. It's one end 'A' is in H.P and 12 mm in front of V.P. Draw the projections of AB and determine its inclination with H.P and V.P. **04**
- (c) A line AB 100 mm long is inclined to H.P at 45° and inclined to V.P at 30°. Draw front and top views of line and determine their lengths. Also determine the perpendicular distance of end B from both H.P and V.P. **07**

OR

- (c) A line PQ 85 mm long has its end 'P' 10 mm above H.P and 15 mm in front of V.P. The top view and front view of line PQ are 75 mm and 80 mm respectively. Draw its projections. Also determine the true and apparent inclinations of the line. **07**
- Q.3** (a) A square lamina of 40 mm side rests on one of its sides on H.P. The lamina makes 30° with H.P and the side on which it rests makes 45° with V.P. Draw its projections. **03**
- (b) Draw the development of the lateral surfaces of a square pyramid, side of base 25 mm and height 50 mm, resting with its base on H.P and an edge of the base is parallel to V.P. **04**
- (c) A hexagonal prism with side of base 30 mm and axis length 60 mm is resting on one of its base edge on H.P such that its axis is inclined at 45° with H.P and the side on which it is resting is inclined at 30° with V.P. Draw the projections. **07**

OR

- Q.3** (a) Give complete classification of solids. **03**
- (b) A regular pentagon laminate of 30 mm each side is resting on H.P on one of its sides with its surface making 45° with H.P. Draw its projection when the side in H.P makes an angle 30° with V.P. **04**

- (c) A square base pyramid with 40 mm side and axis 65 mm long, has its base on the HP and all the edges of the base are equally inclined to the VP. It is cut by a section plane perpendicular to the VP and inclined at 45° to the HP. Further, it is bisecting the axis of the pyramid. Draw its sectional top view, sectional side view and true shape of the section. **07**
- Q.4** (a) Write any three main differences between first angle and third angle projection system. **03**
- (b) In a slider crank mechanism (**Figure-1**), the connecting rod is 160 mm and crank is 40 mm in length. The other end point of connecting rod on the slider moves along a straight line passing through the center of crank rotation. Trace the locus of midpoint 'P' of the connecting rod. **04**
- (c) Draw front view, top view and right-hand side view of the object given in **Figure-2** using first angle projection. **07**
- OR**
- Q.4** (a) Explain following commands: (i) Line (ii) Poly Line (iii) Circle **03**
- (b) List and explain different methods to draw a polygon in AUTOCAD. **04**
- (c) **Figure-3** shows the pictorial view of an object, draw the following views using first angle method of projection. Also give the important dimensions. **07**
- (i) Sectional front view, along section A-A
- (ii) Top View
- Q.5** (a) What are the benefits of Computer Aided Drawing. **03**
- (b) Draw isometric circle on the three side of cube of 60 mm dimension. **04**
- (c) Draw isometric drawing of an object whose projections are given in **Figure-4**. **07**
- OR**
- Q.5** (a) Draw the figure to explain the aligned and unidirectional system of dimensioning. **03**
- (b) Draw an isometric scale of 100 mm length and show 30 and 60 mm length on the scale. **04**
- (c) Draw isometric drawing of an object whose projections are given in **Figure-5**. **07**

FIGURES

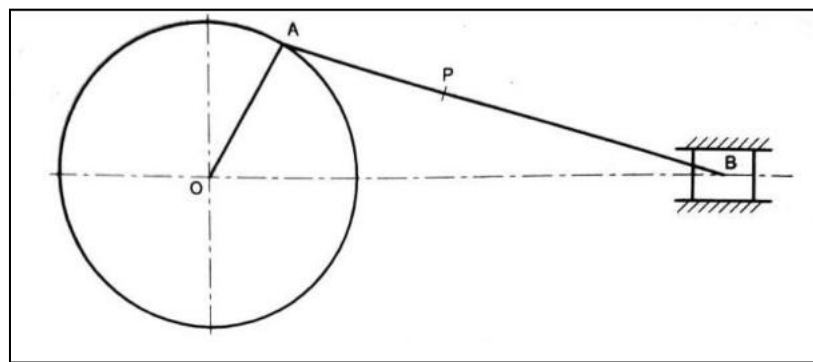
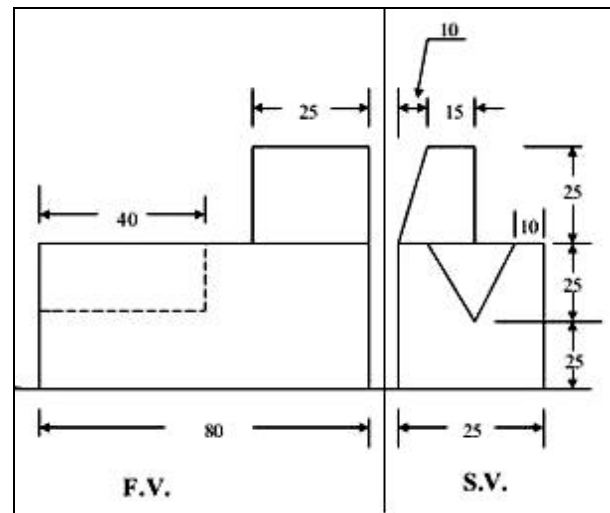
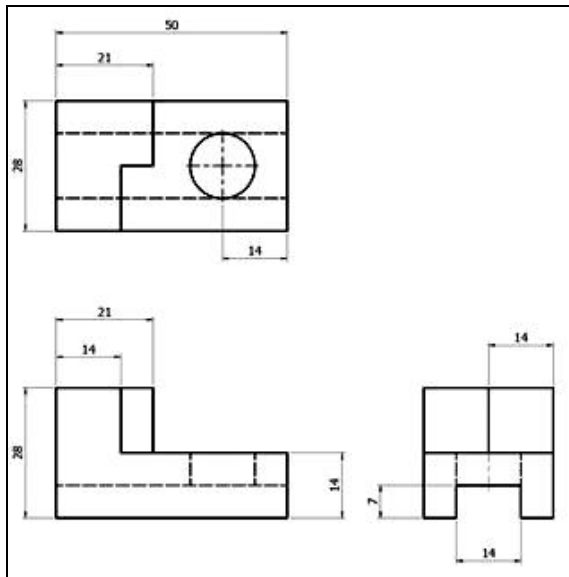
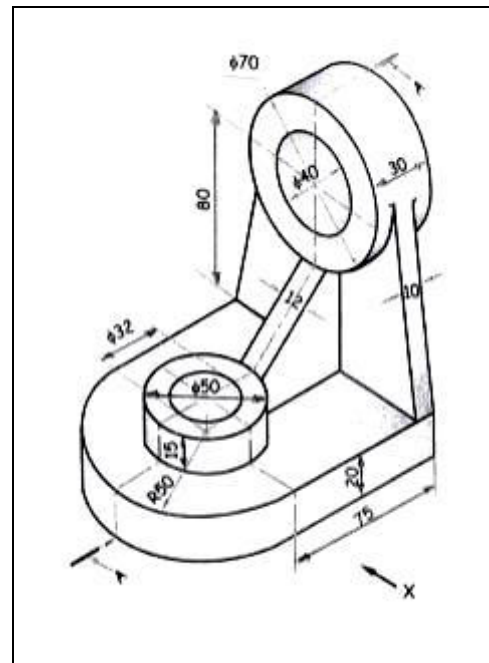
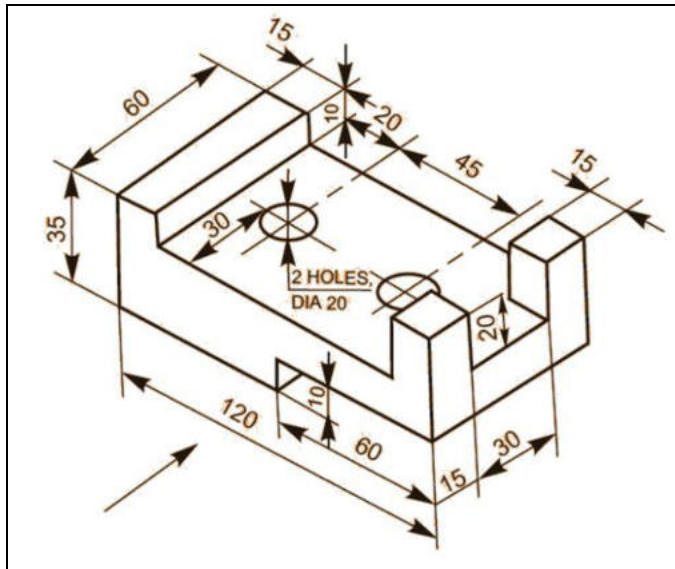


Figure-1[Q4 (b)]



GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110013****Date:28/03/2022****Subject Name:Engineering Graphics & Design****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) (i) Which type of line is used to indicate long break line (continuous thin with zig-zags line)? **3**
(ii) Which type of line is used to indicate hidden lines (dashed medium thick line)?
(iii) What is Representative Fraction/Scale Factor?

- (b) The foci of an ellipse are 110 mm apart. The minor axis is 70 mm long. Determine the length of major axis and draw half ellipse by rectangular method. **4**
(c) Construct the involute of circle of 30 mm diameter for one turn. Draw tangent and normal to the involute at any point on it. **7**

- Q.2** (a) Draw the projections of the following points on the same X-Y line. **3**
(1) Point A 25 mm below H.P. and 20 mm in front of V.P.
(2) Point B 35 mm above H.P. and 40 mm in front of V.P.
(3) Point C on V.P. and 30 mm above H.P.
(b) The foci of an ellipse are 110 mm apart. The minor axis is 70 mm long. Draw half ellipse by concentric circle method. **4**
(c) A circular disc of diameter AB = 90 mm, rotates with uniform angular velocity. The point P which is at A, moves with uniform linear velocity and reaches the point B, When the disc completes one revolution. Trace the locus of point P moving from A to B. **7**

OR

- (c) Construct the involute of a hexagon of side 20 mm. Draw tangent and normal to the involute at any point. **7**

- Q.3** (a) Draw projection of line CD is 50 mm long, when it is not contained by any plane and parallel to both planes. A line CD is 20 mm above H.P. and 25 mm in front of V.P. **3**
(b) A circular plane having 50 mm diameter is resting with point A of its periphery on H.P. The surface of plane is inclined with 30° to H.P. Draw the projection of this circular plane. **4**
(c) A straight line AB 110 mm long is inclined at 30° to H.P. and 45° to V.P. One of its ends A is 40 mm from the V.P. and 20 mm above H.P. Draw the projection of line AB. **7**

OR

- Q.3** (a) A line PQ 130 mm long is parallel to the V.P. and inclined at 60° to H.P. The end nearest to the H.P. is 30 mm from it and 40 mm from the V.P. Draw the projection of line PQ. 3
- (b) A rectangular plane ABCD having 60 mm $\times$ 30 mm size is parallel to H.P. and perpendicular to V.P. and P.P. Draw the projections of the rectangular when it is 40 mm above H.P. and one of the longer sides is parallel to V.P. and 20 mm in front of it. 4
- (c) A square plane ABCD is of 30 mm side. It is kept on V.P. on one of its corner and it is inclined to V.P. at an angle of 30° . The surface of the plane is perpendicular to H.P. Draw the projection of plane 7
- Q.4** (a) Draw the following sketches: Truncated Cylinder, Frustum of a cone and Frustum of a square pyramid. 3
- (b) A square prism edge of base 35 mm and axis length 50 mm is resting on its base on the H.P. with an edge of base parallel to V.P. Draw the projection of prism. 4
- (c) Draw the development of lateral surfaces of a pentagonal prism with edge of base 40 mm and height 90 mm, kept on H.P. on its edge of base with an angle of base parallel to V.P. When it is cut by an AIP inclined at 30° to H.P. and bisecting the axis of the prism. Refer Figure No 1 7

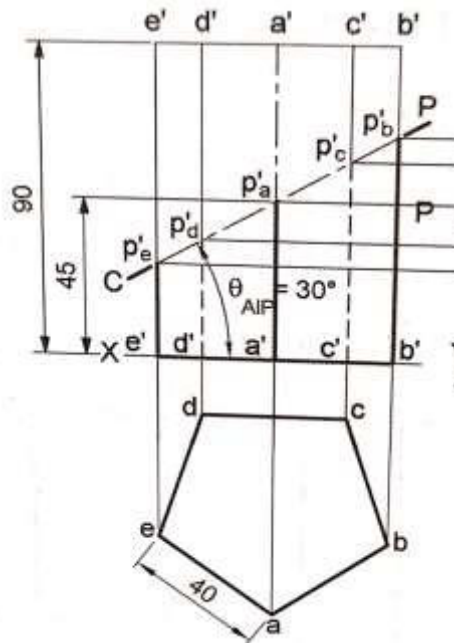


Figure No 1

OR

- Q.4** (a) A square prism side of base 30 mm and its axis is 50 mm is kept on H.P. on its base such that two side of base are perpendicular to V.P. It is cut by the horizontal cutting plane which cuts the axis at its midpoint. Draw front view and sectional top view. 3
- (b) Differentiate isometric projections and orthographic projections. 4

- (c) Figure No. 2 shows pictorial view of an object. Draw Front view looking in the direction F and also draw any one side view. 7

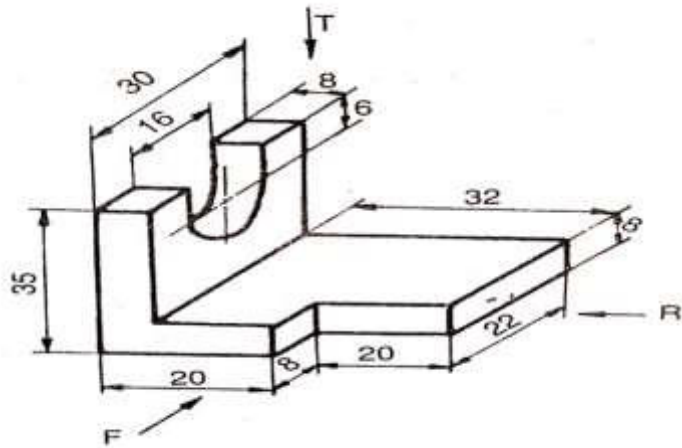


Figure No. 2

- Q.5 (a) Draw the pictorial view of truncated hexagonal pyramid. 3
 (b) What are the various applications of Auto CAD? 4
 (c) Figure No. 3 shows pictorial view of an object. Draw Front view looking in the direction F and also draw any one side view. 7

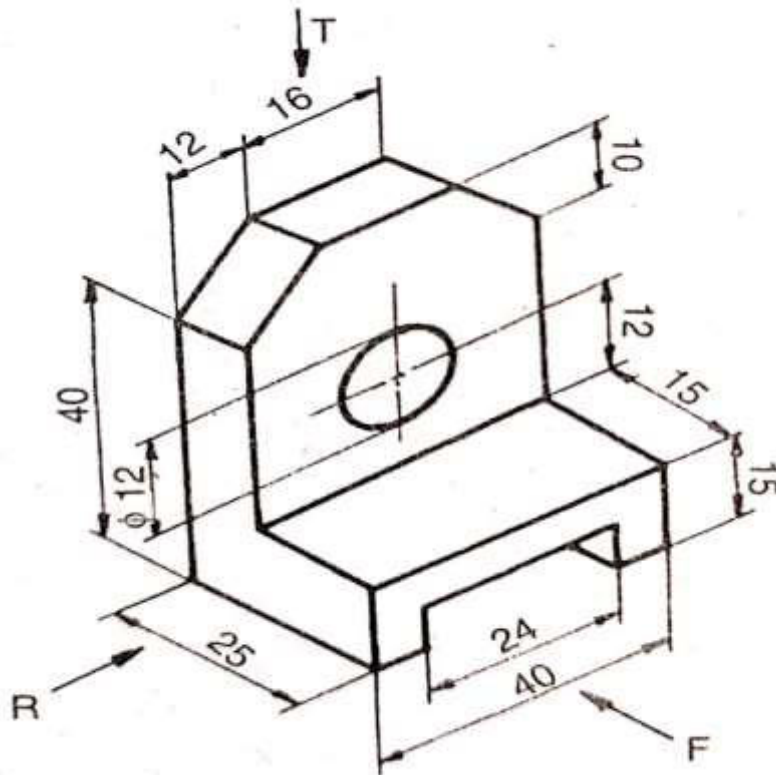


Figure No. 3

OR

- Q.5** (a) State some of important command use in auto CAD. 3
(b) What are the various advantages of Auto CAD? 4
(c) Figure No. 4 shows front view and top view of an object. Draw isometric 7
view. Total height cover by the object is 60 mm.

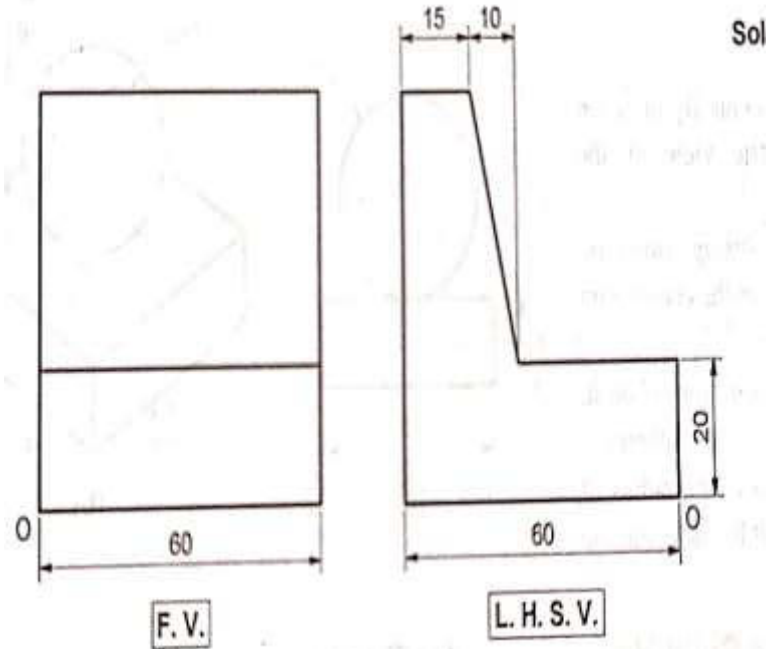


Figure No. 4

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022****Subject Code:3110013****Date:10-08-2022****Subject Name:Engineering Graphics & Design****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Explain Representative Fraction. Construct a plain scale of R.F. = 1:50 to show meters and decimeters and long enough to measure 8 meters. Indicate 5.7 m distance on it.	03
	(b) Find the locus of a point P, moving in a plane, keeping its distances equal from two non-parallel fixed straight lines O_1O_2 and O_1O_3 .	04
	(c) Crank OA rotates about 'O'. (Slide along 'OC', rod BD is fixed to connecting rod AC). Draw locus of point 'D' for one complete rotation of crank OA. (Refer figure 1)	07
Q.2	(a) Construct a diagonal scale of RF = 1/4000 to show 374 meters and long enough to measure up to 500 meters.	03
	(b) The distance between Delhi and Agra is 200 km. In a railway map it is represented by a line 5 cm long. Find it's R.F. Draw a diagonal scale to show single km. And maximum 600 km. Indicate on it the distance of 322 km.	04
	(c) The foci of an ellipse are 110 mm apart. The minor axis is 70 mm long. Determine the length of major axis and draw half ellipse by rectangle method and other half by concentric circle method.	07
OR		
	(c) A circle of 50 mm diameter rolls on the circumference of another circle of 150 mm diameter and outside it. Draw the locus of the point P on the circumference of the rolling circle for one complete revolution of it. Take initial position of point P at the contact point between two circles. Name the curve.	07
Q.3	(a) Draw a cycloid for a rolling circle of 60 mm diameter rolling along a straight line without slipping. Take initial position of the tracing point at the bottom of the vertical centre line of the rolling circle.	03
	(b) Define the following terminology with respect to conics and also show them in a diagram: (i) Focus, (ii) Directrix, (iii) Axis, (iv) Vertex	04

- (c) A point P is 120 mm away from the fixed point pole O. A point P moves towards pole O and reaches the position P' in one convolution where OP' is 22 mm. The point P moves in such a way that its movement towards fixed point O, being uniform with its movement around fixed point pole O. Draw the curve traced out by the point P. Name the curve. **07**

OR

- Q.3 (a)** Define eccentricity ratio. Write the values of eccentricity for (i) ellipse, (ii) parabola, and (iii) hyperbola **03**

- (b) Construct an ellipse when its major axis is 100 mm and minor axis 70 mm by using concentric circle method. **04**

- (c) Construct a parabola, when the distance of the focus from the directrix is 40 mm. Also draw a tangent and normal at a point on it 30 mm from F. **07**

- Q.4 (a)** A rectangular plane ABCD having 60 mm X 30 mm size is parallel to V.P., and perpendicular to H.P., and P.P. Draw the projections of the rectangle when it is 40 mm in front of V.P., and one of the smaller sides is parallel to H.P. and 20 mm above it. **03**

- (b) A line PQ 100 mm long is inclined at an angle of 45° to H.P. and 30° to V.P. One of its end point 'P' is in H.P. as well as VP. Determine its apparent inclination with V.P. **04**

- (c) A square pyramid, side of base 35 mm and axis length 50 mm is lying on the H.P. on one of its triangular faces. Draw the projections of the pyramid when the base edge contained by the triangular face on H.P. makes an angle of 45° on the V.P. keeping apex of the pyramid towards the observer. **07**

OR

- Q.4 (a)** A line PQ 70 mm long is parallel to V.P. and 30° inclined to H.P. The end P is 30 mm above H.P. and 20 mm in front of V.P. Draw the projections. **03**

- (b) A square pyramid, base 45 mm side and axis 70 mm long has its base in H.P. all edges of the base are equally inclined to V.P. It is cut by a section plane perpendicular to V.P. and inclined at 45° to the H.P. such that it bisects the axis. Draw its sectional top view and front view. **04**

- (c) A line AB, 75 mm long is inclined at an angle 35° to the H.P. and 55° to the V.P. Its end point 'A' is on the H.P. and 15 mm in front of the V.P. Draw the projections of the line AB assuming it to be in the first quadrant. **07**

- Q.5 (a)** Write the advantages of CAD. **03**

- (b) Prepare an isometric scale to measure 40 mm and 74 mm. **04**

- (c) Figure 2 shows F.V. and R.H.S.V. of an object. Draw isometric view. **07**

OR

- Q.5 (a)** Write various commands used in AutoCAD and their usage. **03**

- (b) Write the comparison between first angle projection method and third angle projection method. **04**

- (c) Draw orthographic views of the figure 3 by using first angle projection method: **07**

(i) Full sectional front view

(ii) Top view

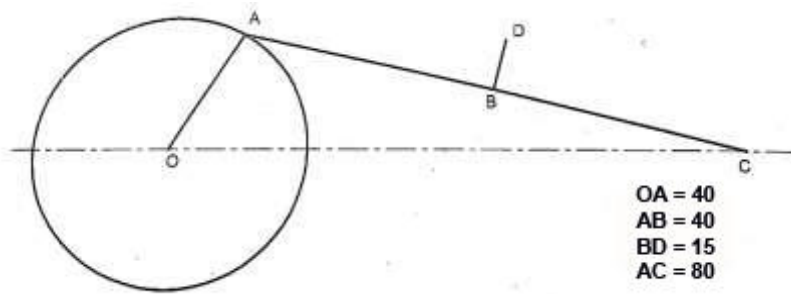


Figure 1

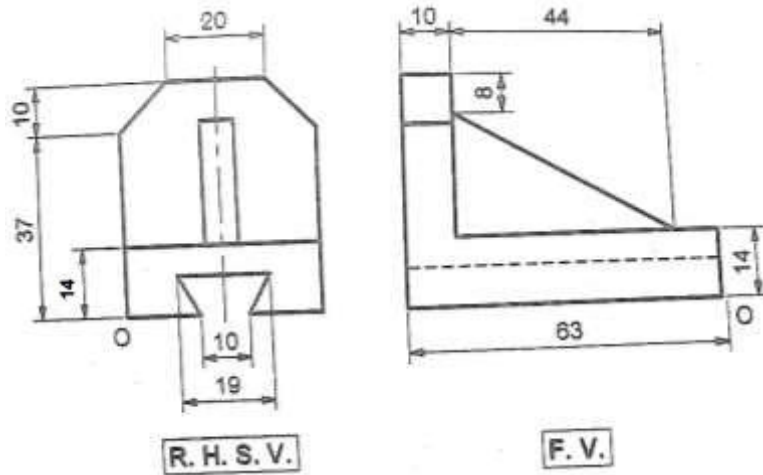


Figure 2

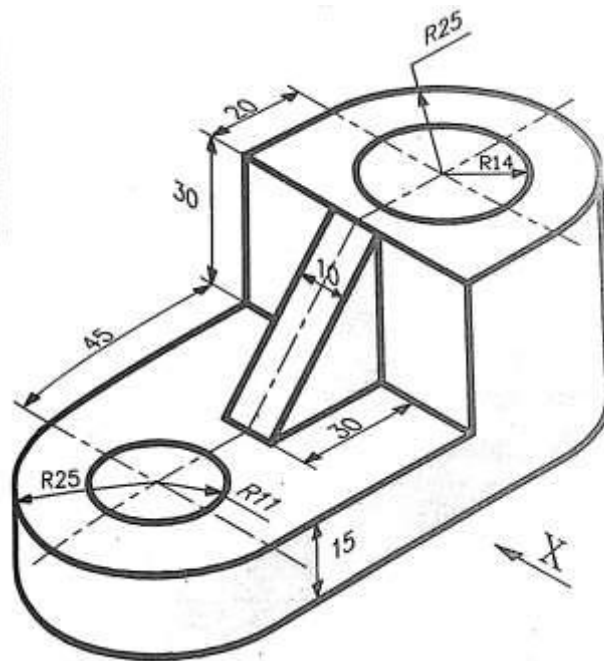


Figure 3

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110013****Date:10-03-2023****Subject Name:Engineering Graphics & Design****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define representative fraction (R.F.) of scale. Also list types of scale according to value of R.F.	03
	(b) In a slider crank mechanism OBA, the crank OB is 200 mm long and the connecting rod BA is 850 mm long. Plot the locus of point P where P is midpoint of connecting rod. Refer figure 1.	04
	(c) Construct one complete turn of an involute of a circle having diameter 30 mm. Also draw tangent and normal to the curve S. The point S is a point on the involute and it is at a distance of 45mm from centre of the circle.	07
Q.2	(a) Draw the projection of following points on the same X-Y line. (1) Point A is 20 mm above HP and 30 mm behind V.P. (2) Point B is on HP and 40 mm in front of V.P. (3) Point C is 10 mm below HP and 25 mm behind V.P.	03
	(b) A line MN, 60 mm long, is in V.P. It makes an angle of 30° with the H.P. The end point M is 15 mm above H.P. Draw projections of line MN. Also measure Plan length of the line.	04
	(c) The line AB has its end A, 15 mm above H.P. and 10 mm in front of V.P. The end B is 60 mm above H.P. The distance between the end projectors is 50 mm. The plan of the line is inclined at 25° to X-Y line. Draw the projections of line. Find true length and inclination of the line with H.P.	07
	OR	
	(c) Draw the projections of a straight line 80 mm long inclined at 60° to H.P. and 30° to V.P. with end A in the H.P. and the end B in the V.P.	07
Q.3	(a) A pentagonal plate, side 25 mm, is inclined to H.P. by 30°. It is perpendicular to V.P. and resting on HP on its one of the side. Draw projections of the plate.	03
	(b) A circular disk, 50 mm in diameter, is resting on the H.P. on one of the point A of the circumference. Plane is inclined to the H.P. such a way that the top view of it is an ellipse of minor axis 40 mm. Draw its projections.	04
	(c) A cone, base diameter 30 mm and height 60 mm, is resting on its base. It is cut by AIP inclined at 45° to H.P. and passing through midpoint of the axis of cone. Draw front view, sectional top view and true shape of this sectional solid.	07
	OR	
Q.3	(a) A hexagonal plate, side 25 mm, is inclined to H.P. by 45°. It is resting on HP on one of the corner. The plate is perpendicular to V.P. Draw projections of the plate.	03

- (b) A square plate, 40 mm side, is resting on the H.P. on one of its sides. The plate is inclined to H.P such a way that it's top view looks like a rectangle with 20 mm shorter side. Draw its projections. **04**
- (c) A square prism, side of base 30 mm and height 45 mm, is resting on H.P. on one of the corners of the base. The longer edge containing that corner on which it rests, is inclined at 45° to H.P. and top view of it is inclined at 30° to V.P. Draw the projections when top end of the prism is nearer to V.P. **07**
- Q.4** (a) Develop the complete surface of a square prism of side of base 40 mm and height 80 mm. **03**
- (b) Draw front view of an object shown in figure 2 using first angle projection methods. **04**
- (c) Draw top view and left hand side view of object shown in figure 2 using first angle projection methods. **07**
- OR**
- Q.4** (a) Draw the development of the lateral surface of a square pyramid, side of base 30 mm and height 50 mm, resting with its base on H.P. All edge of the base are equally inclined to V.P. **03**
- (b) Draw front view of an object shown in figure 3 using first angle projection methods. **04**
- (c) Draw top view and left hand side view of object shown in figure 3 using first angle projection methods. **07**
- Q.5** (a) Explain any three AutoCAD commands in short **03**
(1) Spline (2) Rotate (3) Polygon (4) Multiline
- (b) Draw isometric scale showing 100 mm isometric length equivalent to 100mm normal length. **04**
- (c) Draw isometric projection of an object whose front view and top view are shown in figure 4. **07**
- OR**
- Q.5** (a) Explain any three AutoCAD commands in short. **03**
(1) Circle (2) Array (3) Mirror (4) Fillet Command
- (b) Draw isometric shape of a circle having 60 mm in diameter using four centre methods. **04**
- (c) Draw isometric projection of an object whose front view, top view and side view are shown in figure 5. **07**

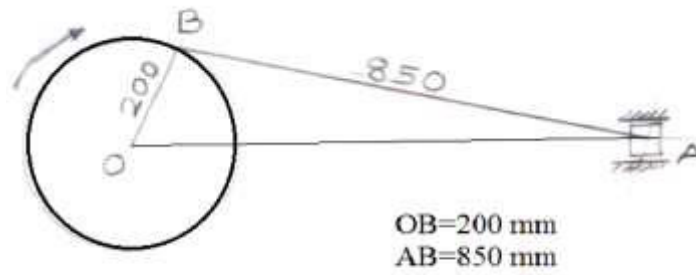


Figure 1

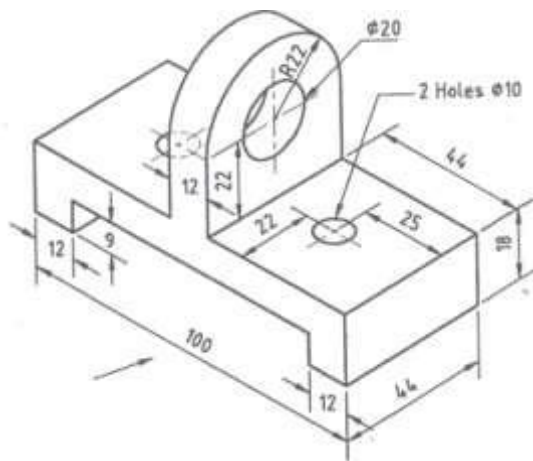


Figure 2

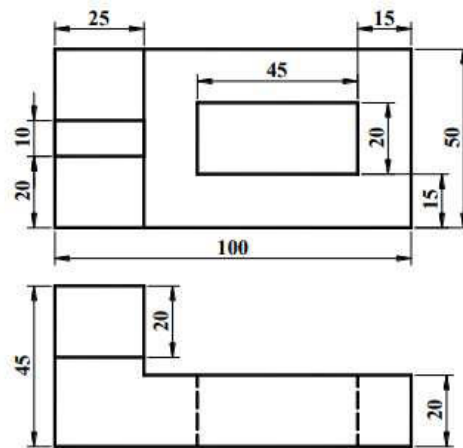


Figure 4

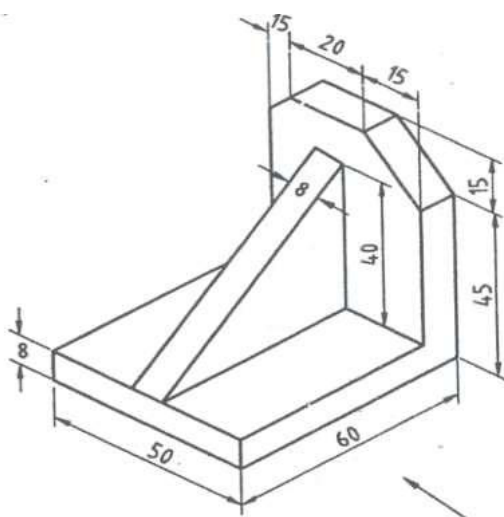


Figure 3

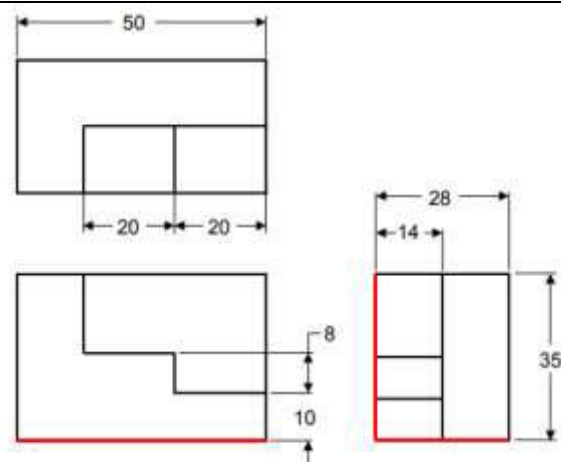


Figure 5

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023

Subject Code:3110013

Date:05-08-2023

Subject Name:Engineering Graphics & Design

Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define the functions with symbols: 1) continuous thin, 2) Chain thin and thick at ends and 3) Chain thin	03
	(b) Construct a scale to measure km, 1/8 km and 1/40 of a km, in which 1 km is represented by 4 cm. Mark on this scale a distance of 2.775 km.	04
	(c) Construct the Involute of hexagon of 25 mm sides.	07
Q.2	(a) Draw the projections of the following points on the same ground lines, keeping the projectors 15 mm apart: (1) A in the H.P. and 20 mm behind V.P. (2) B 25 mm below the H.P. and 25 mm behind V.P. (3) D 40 mm below H.P. and 25 mm in front of V.P.	03
	(b) A person travels in a curved path such that 25 mm distances from a fixed straight line and fixed point remain the same. Construct the traced curvature path by person and name the curve.	04
	(c) A wheel rolls over the horizontal straight road and covers 220 mm distance in one rotation. Draw the path traced by the point P which is initially at the point of contact between the wheel and the road.	07
	OR	
	(c) Draw an Archimedean spiral of 1½ convolutions, the greatest and least radii being 115 mm and 15 mm respectively. Draw a tangent and a normal to the spiral at a point 63 mm from the pole.	07
Q.3	(a) A line CD, 80 mm long is inclined at 45° to H.P. and 30° to the V.P., its end C is in H.P. and 40 mm in front of V.P. Draw the projections.	03
	(b) A hexagonal plane of 30 mm side, rests on the V.P. on an edge such that the surface is inclined at 45° to the V.P. Draw its projections.	04
	(c) A circular plane of 80 mm diameter has one of the ends of the diameter in the H.P. while the other end is in the V.P. The plane is inclined at 30° to the H.P. and 60° to the V.P. Draw its projections.	07
	OR	
Q.3	(a) A 100 mm long line PQ is inclined at 30° to H.P. and 45° to the V.P. Its mid-point is 35 mm above the H.P. and 50 mm in front of V.P. Draw its projections.	03
	(b) The top view of a lamina whose surface is perpendicular to V.P. and inclined at an angle of 45° to H.P. appears as a regular hexagon of 30 mm side, having a side parallel to the reference line. Draw the projections of plane and obtain its true shape.	04
	(c) Draw the projections of a rhombus having 100 mm and 40 mm long diagonals. The bigger diagonal is inclined at 30° to H.P. with one of the end point in H.P. and the smaller diagonal is parallel to both the planes.	07
Q.4	(a) Classify solids with shapes of drawings.	03
	(b) A square pyramid, side of base 40 mm and axis 60 mm is resting on its base on H.P. Draw its projections when (a) a side of the base is parallel to V.P., (b) a side of the base is inclined at 30° to V.P., and (c) all the sides of the base are equally inclined to V.P.	04

- (c) A hexagonal pyramid, 30 mm base side and 60 mm long axis rests on the H.P. with a side of base parallel to V.P. It is cut by planes perpendicular to V.P., to obtain the front view as shown in Fig. 1. Draw the development of the lateral surface of the retained solid.

07

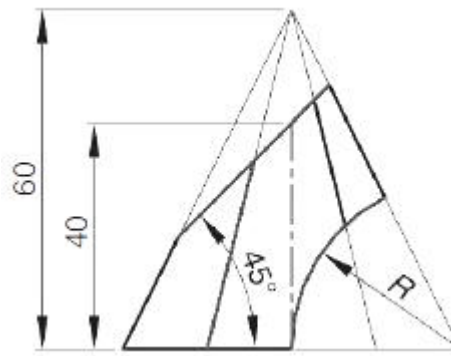


Fig. 1

OR

- Q.4 (a) Explain difference of any two projection methods. (b) A cylinder with 50 mm base diameter and 80 mm long axis, is lying on a generator on the H.P. with its axis parallel to the V.P. It is cut by an A.I.P. inclined at 30° to the H.P. and passes through a point on the axis 30 mm from one of its ends. Draw its anyone sectional top view. (c) Draw any three views using orthographic projections of fig 2.

03

04

07

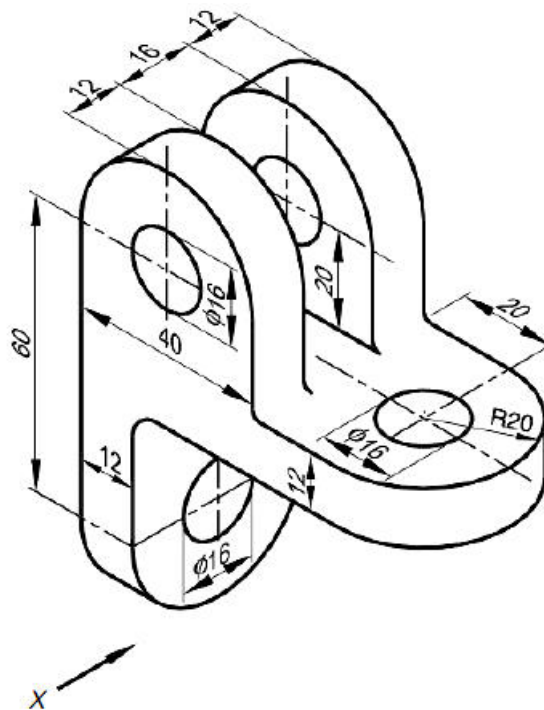


Fig. 2

- Q.5 (a) State the importance of AutoCAD in engineering field. (b) Explain the importance of file extension in the communication of drawing. (c) Draw isometric view of given fig. 3.

03

04

07

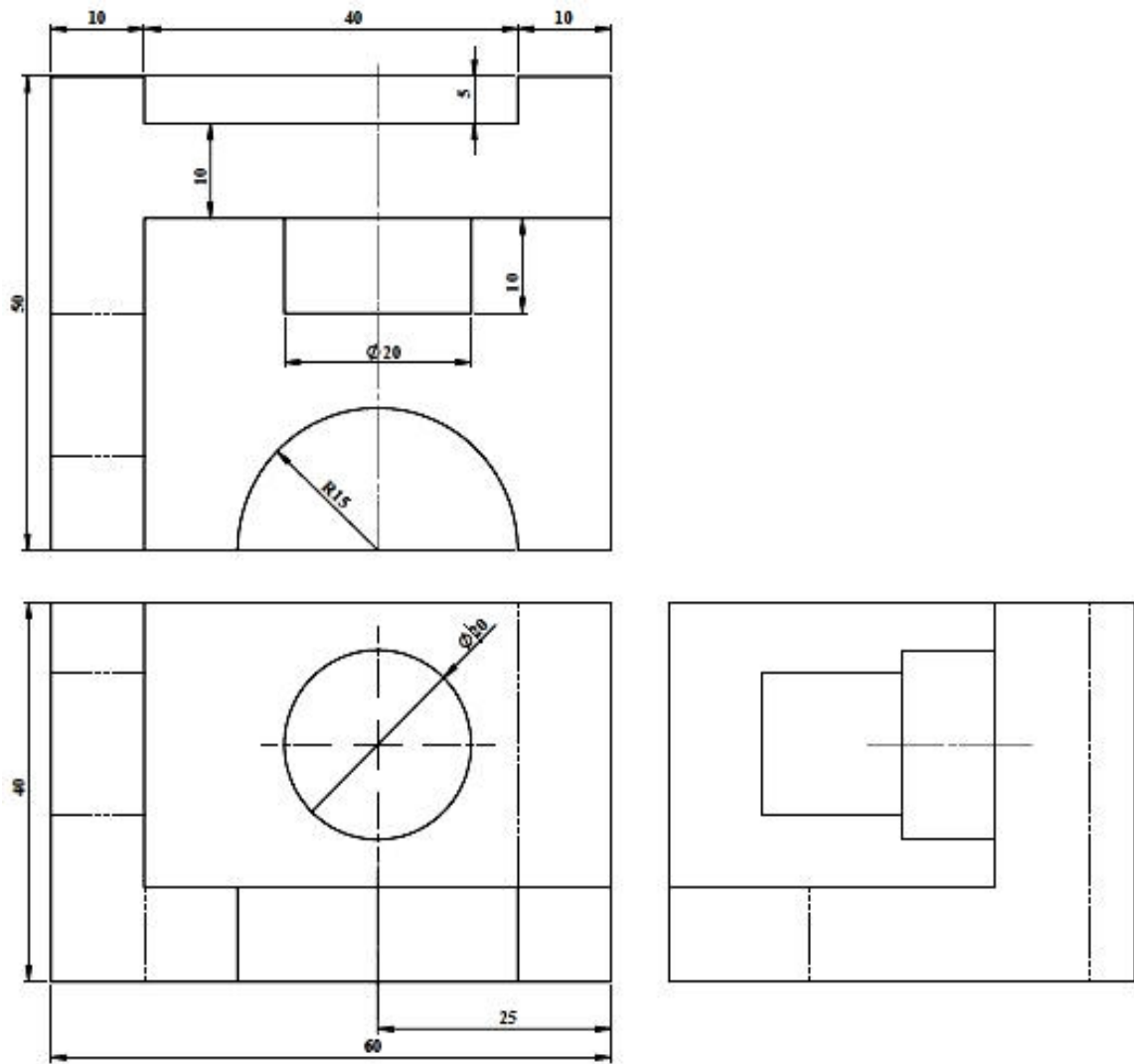


Fig. 3
OR

- Q.5 (a) Enlist any six CAD drawing preparatory commands with its symbols and function. 03
- (b) Identify the command, that used 1) to remove certain unwanted part, 2) to show sectional view, 04
3) to make a continuous line and 4) to make a circular shape at the edge of drawing.
- (c) Draw the Front view of Fig. 4 looking in the X direction and also draw any one side view. 07

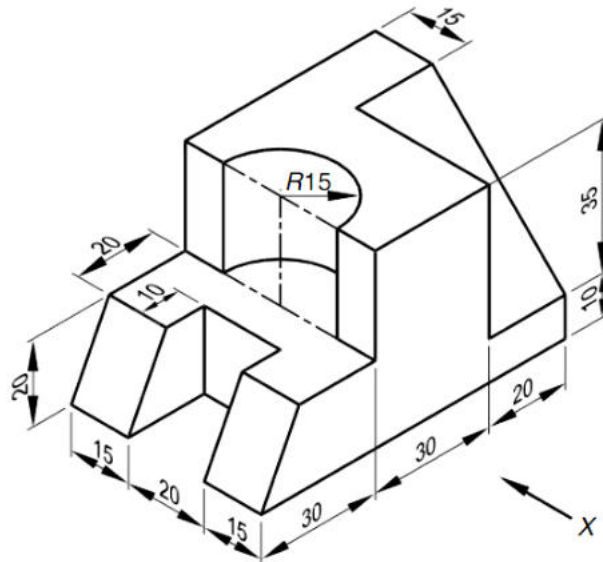


Fig. 4

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023

Subject Code:3110013

Date:19-01-2024

Subject Name:Engineering Graphics & Design

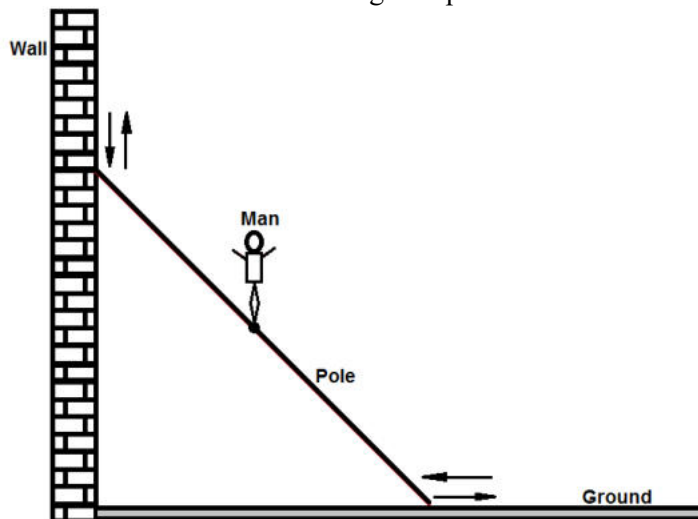
Time:02:30 PM TO 05:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | Marks |
|---|-------|
| Q.1 (a) Define Representative Fraction. With reference to Representative Fraction, classify the scales. | 03 |
| (b) Construct a scale of 1:40 to read meters and decimeters and long enough to measure 6 m. Mark on it a distance of 4.7 m and 3.2 m. | 04 |
| (c) A pole is sliding vertical to horizontal position as shown in figure. A man is standing at the midpoint of pole. Trace locus of man for sliding of the pole from vertical to horizontal. Length of pole is 480 centimeters. | 07 |



- | | | |
|---------|--|----|
| Q.2 (a) | <p>The diagram shows a parabola opening to the right. A vertical line labeled 'D' represents the directrix. A horizontal line represents the axis of symmetry. A point 'P' is on the parabola. A horizontal line segment 'PR' connects 'P' to the directrix 'D'. A line segment 'PF' connects 'P' to the focus 'F' on the axis. The focus 'F' is at a distance from the vertex equal to the distance from the vertex to the directrix.</p> | 03 |
|---------|--|----|

In the figure shown here, show the conditions for having Ellipse, Parabola and Hyperbola for the following equation

$$\frac{PF}{PR}$$

- | | |
|---|----|
| (b) Identify the following types of line and write their use in engineering drawing | 04 |
| 1) | |

2) 

3) 

4) 

- (c) A thin circular disc of 50 mm diameter is allowed to roll without slipping from upper edge of sloping board which is inclined at 15° with the horizontal plane. Draw the curve traced by the point on the circumference of the disc. Consider the point is at contact surface of disc and sloping board initially. 07

OR

- (c) Major axis of an ellipse is 120 mm and the distance between foci of ellipse is 80 mm. Draw an ellipse using oblong method. Draw tangent and normal to any point to the curve. 07

Q.3 (a) Differentiate between prism and pyramid 03

- (b) A square ABCD of 50 mm side is resting on HP on one of its side such that it appears as a rectangle of 35 mm X 50 mm. Draw its projection and find its inclination with HP. 04

- (c) A semicircular plate has the 60 mm diameter straight edge in HP and inclined at 45° to VP. The surface of plate makes an angle of 30° to HP. Draw its projection. 07

OR

Q.3 (a) Show by sketches the difference between
(i) aligned dimensioning and
(ii) unidirectional dimensioning 03

- (b) A line PQ 70 mm long is parallel to profile plane and inclined at 30° to HP. Point P is 10mm Above HP and point Q is in VP. Draw its projections. 04

- (c) A line AB, 80mm long is inclined to HP at 30° and to VP at 45° . The end A is in HP and VP. Draw projection of line AB. Find length of top view and front view of line and inclination of them with HP and VP. 07

Q.4 (a) Define Parabola, Ellipse and Hyperbola 03

- (b) Draw projection of following points:
1. Point A is in HP and in VP
2. Point B is in VP and 25 mm below HP
3. Point C is 10 mm below HP and 20 mm in front of VP
4. Point D is 20 mm above HP and 20 mm in front of VP 04

- (c) A pentagonal pyramid, side of base 25 mm and height 50 mm is resting on HP with one of the edge of base is perpendicular to VP. It is cut by auxiliary inclined plane inclined at 45° and cuts the axis at 20mm above the base. Draw sectional top view, front view and true shape of section. 07

OR

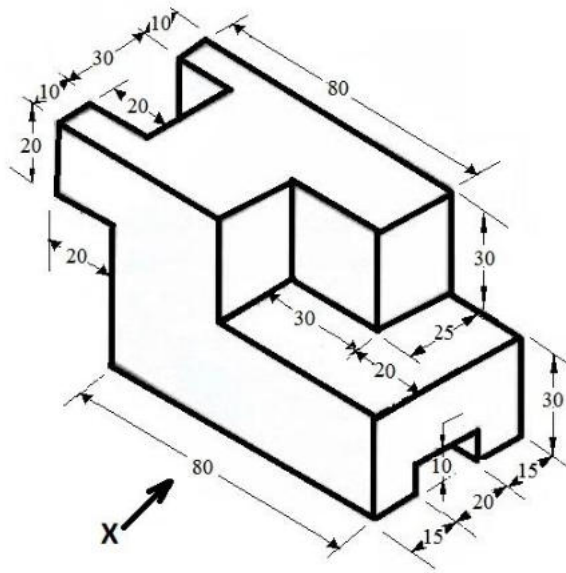
Q.4 (a) Benefits of AUTOCAD over conventional drawing techniques 03

- (b) Hexagonal pyramid of side of base 25 mm and height 55mm is resting on HP on its base. It is cut by a plane parallel to HP and perpendicular to VP and cutting the axis at 25 mm above the base. Draw its projection. 04

- (c) A $30^\circ - 60^\circ$ set square of 60 mm longest side is so kept such that the longest side is in HP making an angle of 30° with VP. The set square itself is inclined at 45° to HP. Draw the projections of the set square. 07

Q.5 (a) What are the advantages and disadvantages of using AUTOCAD? 04

- (b) For the figure shown below, draw plan, elevation and right hand side view using third angle projection method. 10



OR

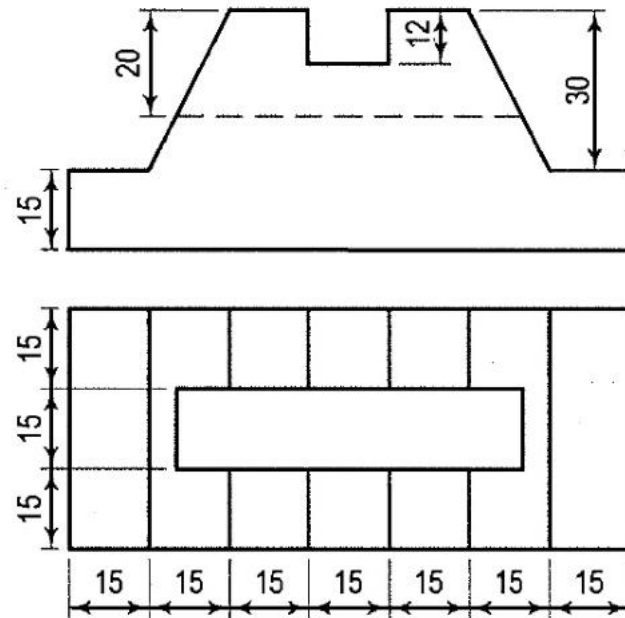
Q.5 (a) Explain use of following AUTOCAD commands

04

1. TRIM
2. ARC
3. MIRROR
4. GRID

(b) Draw isometric view of the following figure

10



GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110013

Date:01-07-2024

Subject Name:Engineering Graphics & Design

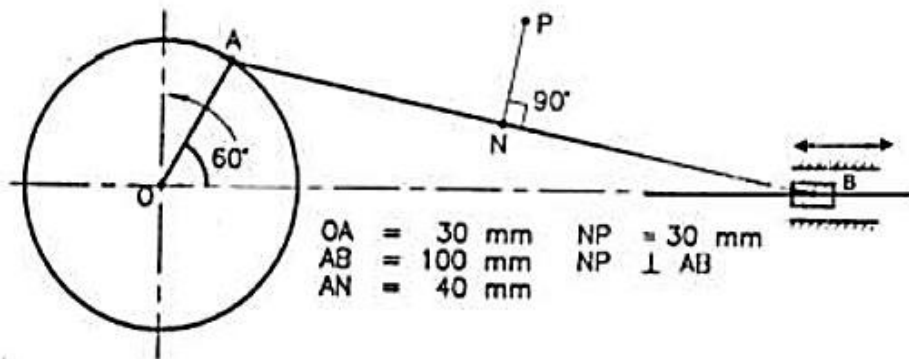
Time:02:30 PM TO 05:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | MARKS |
|---|-----------|
| Q.1 (a) Draw regular square, pentagon and hexagon by UNIVERSAL METHOD of 35 mm side. | 03 |
| (b) The distance between Ahmedabad and Surat is 200 km. In a railway map it is represented by a line 5 cm long. Find it's R.F. Draw a diagonal scale to show single km. And maximum 600 km. Indicate on it following distances. (1) 222 km (2) 569 km. | 04 |
| (c) A pole is of a shape of half hexagon and semicircle. A string is to be wound having length equal to the pole perimeter Draw path of free end P of string when wound completely.
(Take hexagon 30 mm sides and semicircle of 60 mm diameter.) | 07 |
| Q.2 (a) (1) The top view of a right cylinder resting on HP on its base is _____
(2) The development of cylinder is a _____
(3) The isometric projection of a sphere is a _____ | 03 |
| (b) Draw Development of lateral surfaces of following solids.
(1) Cylinder (2) Cone (3) Pentagonal Prism (4) Cube | 04 |
| (c) Rod AB, 100 mm long, revolves in clockwise direction for one revolution. Meanwhile point P, initially on A starts moving towards B and reaches B. Draw locus of point P. | 07 |
| OR | |
| (c) Draw loci of point 'P' given in figure | 07 |

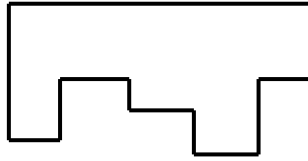


- | | |
|--|-----------|
| Q.3 (a) Draw following projection of points: | 03 |
| (1) Point 'A' is 20 mm above HP and 25 mm Infront of VP. | |
| (2) Point 'B' is 35 mm behind VP and 30 mm above HP. | |
| (3) Point 'C' is in the HP and 15 mm Infront of VP. | |

- (b) Draw projections of line AB=80 mm inclined to 60° HP & 30° VP and parallel to PP. 04
- (c) Line AB 75mm long makes 45° inclinations with VP while it's FV makes 55° . End A is 10 mm above HP and 15 mm in front of VP. If line is in 1st quadrant draw its projections and find it's inclination with HP. 07

OR

- Q.3 (a) Draw Parallel, Chain and Combined Dimensioning for following figure. 03

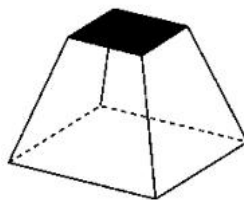
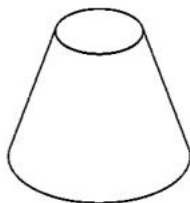


- (b) Write at least one use of following drawing instruments: 04
- | | |
|------------------|------------------|
| (1) Mini drafter | (5) Divider |
| (2) Protector | (6) Roll-N-Draw |
| (3) Set squares | (7) T-square |
| (4) Compass | (8) French curve |
- (c) A room is of size 6.5m L ,5m D,3.5m high. An electric bulb hangs 1m below the center of ceiling. A switch is placed in one of the corners of the room, 1.5m above the flooring. Draw the projections an determine real distance between the bulb and switch. 07

- Q.4 (a) Draw symbol of first angle and third angle projection method. 03
- (b) Draw Isometric view of Circle of 50mm diameter, Pentagon of 30 mm side, a cube of 50mm side and a hexagon of 25 mm side. 04
- (c) A regular pentagon of 30 mm sides is resting on HP on one of its sides with its surface 45° inclined to HP. Draw its projections when the side in HP makes 30° angles with VP. 07

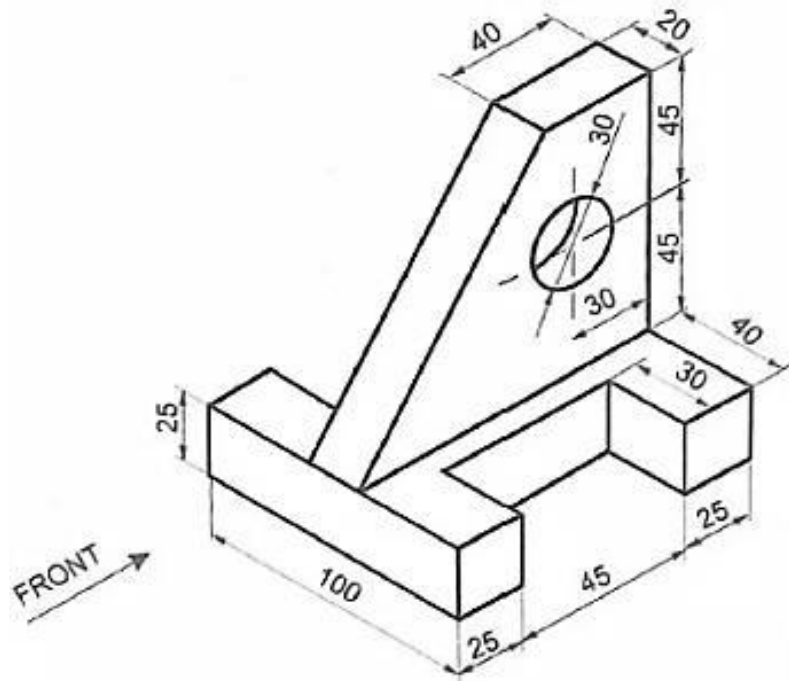
OR

- Q.4 (a) Identify following solids: 03
- (1) (2) (3)



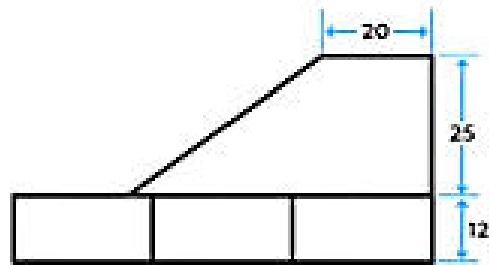
- (b) A rectangle ABCD 60 mm x 40 mm, is parallel to HP with one of its sides inclined at 30° to VP and the end of the side near to VP is 15 mm in front of the VP and 30 mm above HP. Draw its projections. 04
- (c) A square pyramid, side of base 30 mm and height 50 mm, is resting on HP with a side of its base. Draw the projections if the axis is inclined to 45° to HP and parallel to VP 07

- Q.5 (a) What is meant by ARRAY in AUTOCAD? What are types of ARRAYs? 04
- (b) Draw orthographic projections of given figure (FV, TV, RHSV) as per first angle projection method. 10

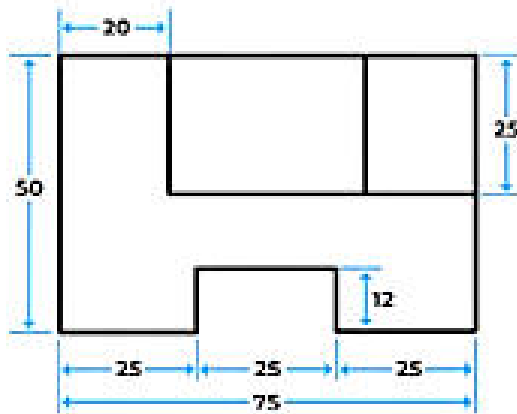


OR

- Q.5 (a) Write down different software used in design. 04
 (b) Draw isometric scale for length of 100mm. and draw isometric view for given figure. 10



Front View



Top View

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:3110013

Date:09-01-2025

Subject Name:Engineering Graphics & Design

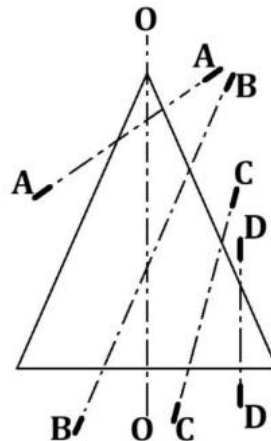
Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

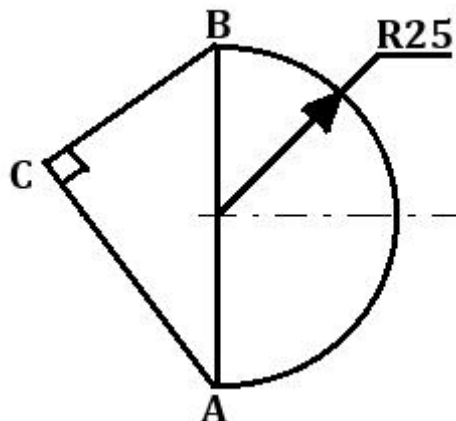
1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Explain reducing scale. Give two practical applications of it.	03
(b) Draw symbol of following and right use of it.	04
1) Cutting Plane line	
2) Dimension line	
3) Break line	
4) Section Line	
(c) The distance between two trees on road is 876m. Show it on a scale of 1:6250. Also show distance of 653m distance between two vehicles parked on a side of the road.	07
Q.2 (a) Define ellipse. What is eccentricity? What is the value of eccentricity for ellipse?	03
(b)	04



When we cut a cone using cutting planes as shown in figure, name the curves which are available as cut section of cone with particular cutting plane.

(c) 07



Draw involute of a string unwound for the given figure from point C, where AC = 40 mm and BC = 30 mm

OR

- (c) Draw an ellipse using concentric circle method. Given the distance between Foci is 90mm and length of major axis is 130mm. **07**
- Q.3** (a) A circular plane of 50 mm diameter is perpendicular to HP and VP. The centre of plane is 30mm above HP and 35mm in-front of VP. Draw its projection. **03**
- (b) A thin rectangular plate of side 40mm x 20mm has its shorter side in HP. Project its front view when its top view is a perfect square of 20mm side. **04**
- (c) The top view of a 75mm long line AB measures 65mm, while its front view measures 50mm. It's one end A is in HP and 12mm in front of VP. Draw the projections of AB and determine its inclination with HP and VP **07**

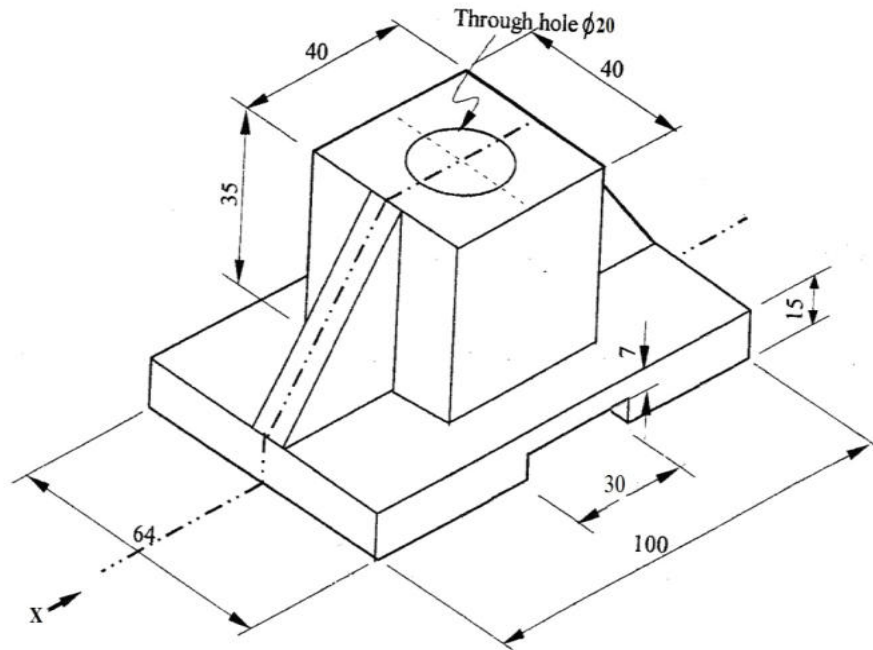
OR

- Q.3** (a) Draw the projection of following points **03**
1. Point A is 20 mm above HP and 35 mm behind VP
 2. Point B is 20 mm below HP and 35 mm in front of VP
 3. Point C is in VP and 15 mm above HP
- (b) A rhombus of 30mm and 50mm diagonal is resting on HP on one of its corner such that it appears a square of 30mm diagonals. Find its inclination with HP. **04**
- (c) A pentagonal plane having edges 25 mm is placed on one of its corners on HP such that the surface makes an angle 30° with HP and side opposite to that corner on which the plane rests inclined at 60° to VP. Draw the top and front views of the plane. **07**
- Q.4** (a) Define solid of revolution? Give three example of solid of revolution. **03**
- (b) A hexagonal prism, edge of base 25 mm and axis 65 mm long, rests with its base on H.P such that one of its rectangular faces is parallel to V.P. It is cut by a plane perpendicular to V.P and parallel to HP and 30mm above HP. Draw the development of the lateral surface of the truncated prism. **04**
- (c) Draw the top view and front view of a rectangular pyramid of sides of base 40 mm x 50 mm and height 70mm when it lies on one of its larger triangular faces on HP. The longer edge of the base of the triangular face lying on HP is inclined at 60° to VP in the top view with the apex of the pyramid being nearer to VP. **07**

OR

- Q.4** (a) Differentiate between prism and pyramid. **03**
- (b) Bisect one of the interior angle of regular pentagonal plane having sides of 30mm. **04**
- (c) A square pyramid having 30mm edge of the base and axis of 70mm long and axis parallel to VP and inclined at 60° to HP. Draw projection of pyramid if one of the edge of its base inclined at 30° to VP and apex is on HP and 40 mm away from VP. **07**
- Q.5** (a) Draw isometric scale for the line of length 100mm. Mark 53mm on it. **04**

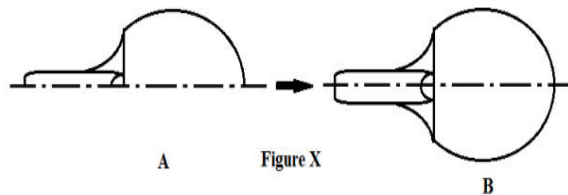
- (b) Draw Front view, Top View and Right hand side view using 1st angle projection method. 10



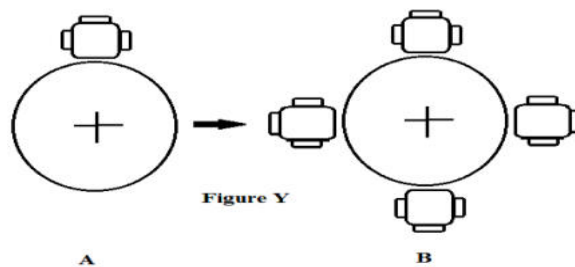
OR

- Q.5** (a) Which AUTOCAD command can be used to create Part B from Part A for the following figures? 03

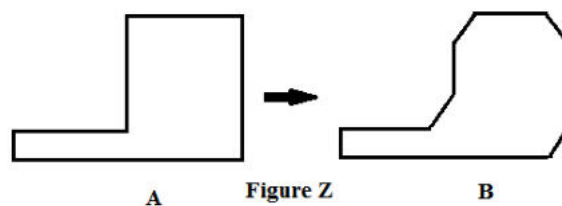
1)



2)



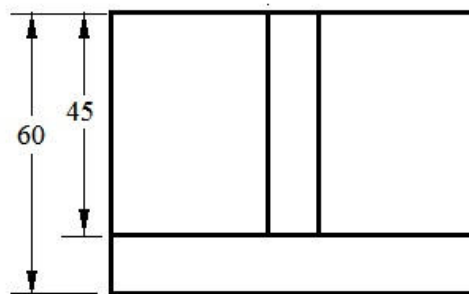
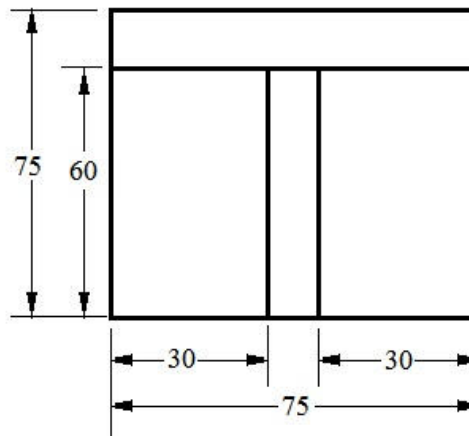
3)



- (b) Advantages of AUTOCAD over conventional drawing. 04

(c) Draw isometric drawing of the following views.

07



GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:3110013****Date:03-07-2025****Subject Name:Engineering Graphics & Design****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Explain the difference between I st angle and III rd angle projection method.	03
	(b) The distance between two cities is 250 km and is represented by a line of length 50mm on a map. Construct a scale to read 600 km and indicate a distance of 530 km on it.	04
	(c) The foci of an ellipse are 120mm apart. The minor axis 72 mm long. Determine the length of the major axis and draw ellipse by oblong method.	07
Q.2	(a) Differentiate in between Unidirectional system and Aligned system of dimensioning.	03
	(b) Draw the projection of following points on the same X-Y line.	04
	(1) Point A is 25 mm above HP and 30 mm behind V.P.	
	(2) Point B is 20 mm above HP and on V.P.	
	(3) Point C is 10 mm bellow HP and 25 mm behind V.P.	
	(4) Point D is on HP and 40 mm in front of V.P.	
	(c) A Line AB, 90 mm long, is inclined to H.P. by 30° and inclined to V.P. by 45°. The line is in first quadrant with Point A 15 mm above H.P. and 25 mm in front of V.P. Draw the projection of line AB.	07
	OR	
	(c) Draw a parabola by rectangle method, given its base as 90 mm and the height of axis is 75 mm.	07
Q.3	(a) Define:	03
	(1) Involute (2) Spiral (3) Cycloid	
	(b) A line AB 75mm long is incline at an angle of 45° to H.P. and 30° to V.P. One of its end point A is H.P. as well as in V.P. Determine its apparent angle with V. P.	04
	(c) In a slider crank mechanism, crank is 40mm and connecting rod is 160mm in length. The slider moves along a straight line passing through center of crank rotation. Trace the locus of midpoint P of connecting rod.	07
	OR	
Q.3	(a) A line PQ 65 mm long is parallel to HP and 40mm below HP Its two ends P and Q are 25mm and 45mm behind VP respectively. Draw the Projections and find inclination of line with VP.	03
	(b) Construct the involute of circle of 30mm diameter for one turn.	04
	(c) A line EF, 75 mm long, has its end E 20 mm below HP and 25 mm behind VP. The end F is 50 mm below H.P. and 65 mm behind VP Draw the projection of line EF and Find its inclination with HP and VP.	07
Q.4	(a) Draw the lateral development of a cone of base diameter 40mm and height 60mm.	03
	(b) A circular plate of diameter 50 mm is resting on HP on a point on the	04

circumference with its surface inclined at 45° to HP and perpendicular to VP. Draw its projections.

- (c) A cone having base diameter 80mm and height 90mm is resting with its base on H.P.. It is cut by A.I.P. at 45° to the H.P. The cutting plane passes through the midpoint of the axis of the cone. Draw the elevation, the sectional plan and true shape of the section. **07**

OR

- Q.4** (a) A square pyramid, base 40 mm side and axis 65 mm long, has its base on the HP and all the edges of the base equally inclined to the VP. It is cut by a section plane, perpendicular to the VP, inclined at 45° to the HP and bisecting the axis. Draw its development. **03**
- (b) An isosceles triangular XYZ having its base XY=50 and altitude 60mm is resting on H.P. on its base XY with its surface making an angle of 45° to H.P. Draw projection of plane. **04**
- (c) A Hexagonal prism is resting on one of its side of base 30mm, such that axis 60mm is inclined at 45° to the HP and the side on which it is resting is inclined at 30° to V.P. Draw the projections. **07**
- Q.5** (a) Explain use of following AUTOCAD commands: **04**
 (1) MOVE (2) FILLET (3) OFFSET (4) CHAMFER
- (b) Draw isometric drawing for the Figure-1. **10**

OR

- Q.5** (a) Discuss advantages of AUTOCAD over conventional drawing. **04**
- (b) Draw the following views for Figure-2 using first angle projection method. **10**
 (1) Sectional Elevation along A-B from X (2) Top view

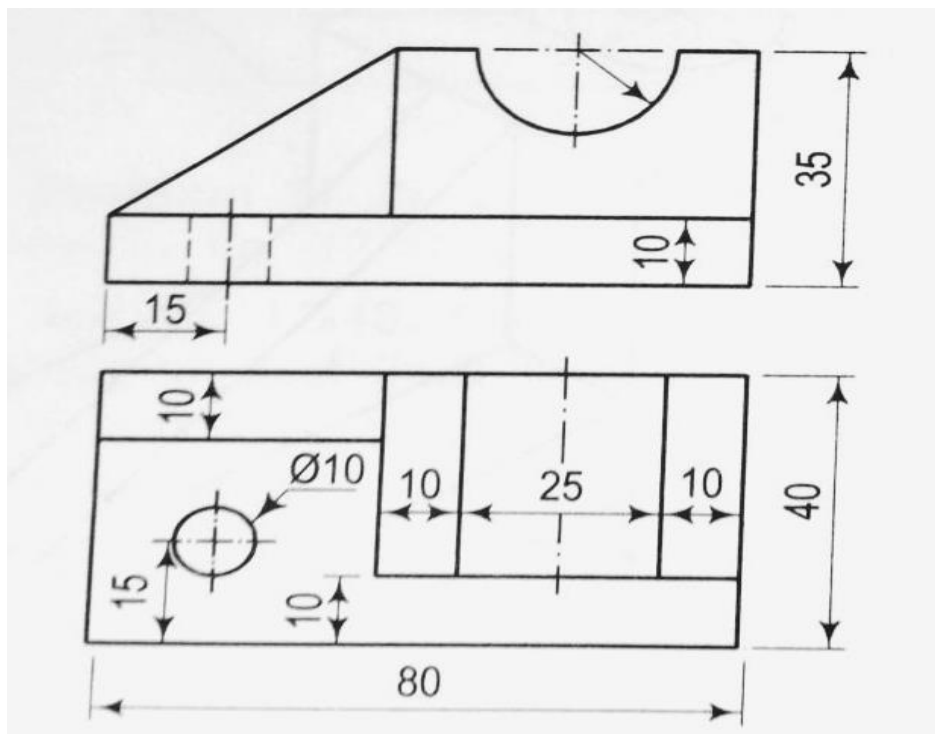


Figure-1

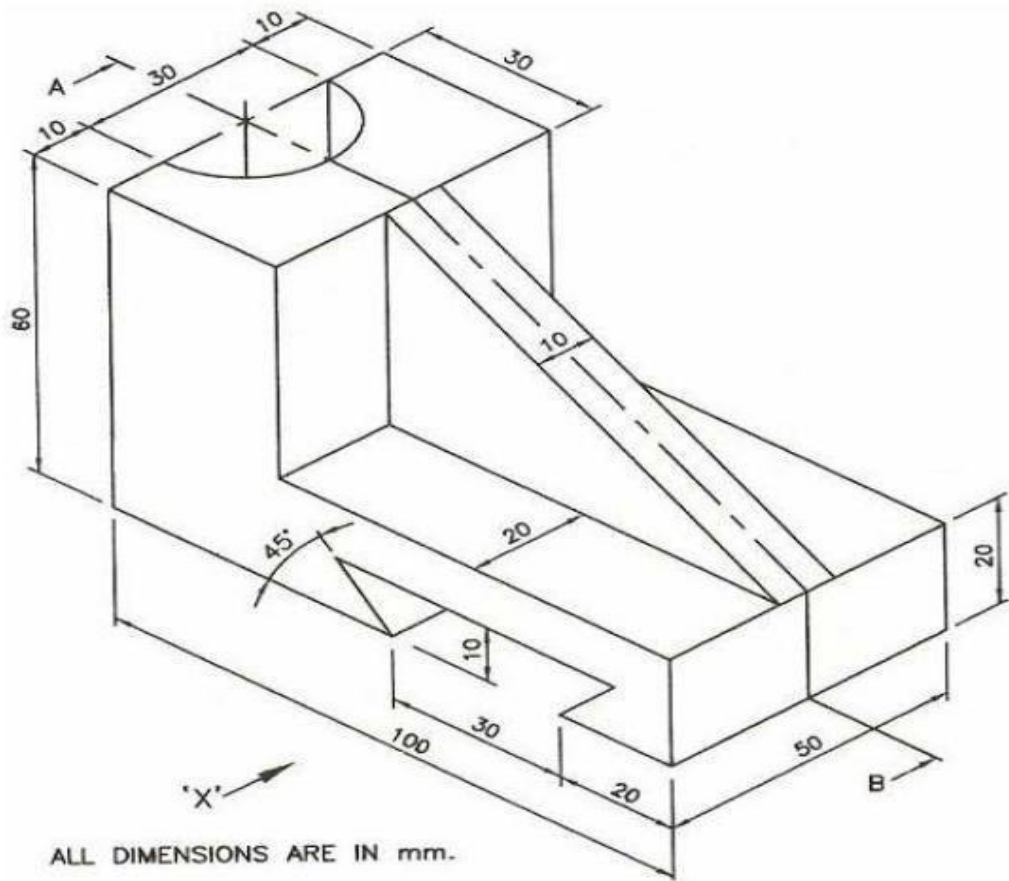


Figure-2

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2025****Subject Code:3110013****Date:05-01-2026****Subject Name: Engineering Graphics & Design****Time:02:30 PM TO 05:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Construct a plain scale of R.F. = 1:10 showing the feet and inches and long enough to read the distance of 5 feet. Show on it the distance of 4 feet and 10 inches.	03
	(b) A circular disc of diameter AB = 90 mm, rotates with uniform angular velocity. The point P which is at A, moves with uniform linear velocity and reaches the point B, when the disc completes one revolution. Trace the locus of point P moving from A to B.	04
	(c) Construct an ellipse when the distance of its focus from its directrix is equal to 50 mm and the eccentricity is $\frac{2}{3}$. Also, draw the tangent and normal to the ellipse at a point 70 mm away from the directrix.	07
Q.2	(a) Draw the projections of the following points on X-Y line: (1) A point 'A' 40 mm below H.P. and 40 mm in front of V.P. (2) A point 'B' 35 mm above H.P. and 45 mm in front of V.P. (3) A point 'C' on V.P. and 30 mm above H.P.	03
	(b) Classify solids based on their geometric properties.	04
	(c) A straight line AB has its end A 10 mm above H.P. and end B 50 mm in front of the V.P. Draw the projections of line AB, if it is inclined to H.P. by 30° and to V.P. by 45° and the true length of the line is 50 mm.	07
	OR	
	(c) The projections of the ends of a line AB are 50 mm apart. The end A is 20 mm above H.P. and 30 mm in front of the V.P. The end B is 10 mm below H.P. and 40 mm behind V.P. Determine the true length of the line AB and its inclination with H.P.	07
Q.3	(a) Explain A.I.P. and A.V.P.	03
	(b) A square prism side of base 35 mm and height 50 mm, is resting on its base in the V.P. and axis perpendicular to the V.P. and 30 mm above the H.P. One of the edges of the base is inclined to 30° to the H.P. Draw the projection of the square prism.	04
	(c) A rhombus ABCD has its diagonal AC = 80 mm and BD = 50 mm. The side AD of the plane is in H.P. and the plane makes an angle of 30° with the H.P. Obtain the projections of the plane when the side CD is in V.P.	07
	OR	
Q.3	(a) What is the apparent shape and the true shape of the section in case of the section of solid?	03
	(b) OBA is a simple slider crank chain. OB is a crank of 30 mm length. BA is a connecting rod of 90 mm length. Slider A is sliding on a straight path passing through point O. Draw the locus of the mid-point of the connecting rod AB for one complete revolution of the crank OB.	04

